Assignment Handout

Competitive Data-Center Connection Planning Game

Course name

October 8, 2026

Assignment in one sentence. You operate a data-center, choose where and how much load to connect in a power grid, optionally build private generation and storage, and maximize your score across three weekly matches.

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1 Assignment overview

Each student group represents a data-center operator such as a cloud-computing company. Your group submits a connection-and-resource plan. Submissions of all groups will be collected and evaluated together using a DC optimal power flow (`DCOPF`) dispatch model. **Groups are ranked by score.**

The classroom game has three weeks and six half-year submissions. Each week has two submissions: Submission 1 is a practice plan and Submission 2 is the scored plan for Weeks 2 and 3. Week 1 is feedback-only and is not included in the final ranking.

1.1 What you must do

Your team must design and submit one complete half-year connection plan in the released CSV format. The goal is to serve valuable computational load while controlling energy, interconnection, generation, and storage costs.

1.2 Deliverables

1. one `UTF-8 CSV` file for your team;
2. validator output showing that the canonical file passes the checker;
3. a short strategy note explaining your main bus, load, generator, and storage choices and the trade-offs you considered;
4. the exact final CSV and any code used to generate it.

The instructor will announce the file name, submission location, deadline, round assignment, and whether the strategy note is required for grading. Do not submit a screenshot in place of the CSV.

1.3 Assignment rules at a glance

- Submit one canonical site-level CSV row for each selected site. The CSV has no hourly period, load, or horizon rows.
- Use `bus IDs` from the supplied ACTIVSg2000 scenario block. Site capacities are specified in MW/MWh.
- The released calendar still determines the evaluator's half-year hourly scenario: 2028 has 4,368/4,416 hours and 2027/2029 have 4,344/4,416 hours for January–June/July–December. Those hours are not CSV rows.
- `Generators` and `storage` are private to the submitting team.
- The active contract has no student-controlled `curtailment` decision.
- The instructor evaluates valid teams jointly and ranks teams by `score`.

Use of AI. You are explicitly encouraged to use available AI tools while doing the assignment, including ChatGPT, Codex, Claude Code, GitHub Copilot, and similar tools. You remain responsible for checking generated code, understanding your submission, respecting the CSV contract, and explaining your engineering decisions. AI assistance is part of the intended computational workflow, not a substitute for validation.

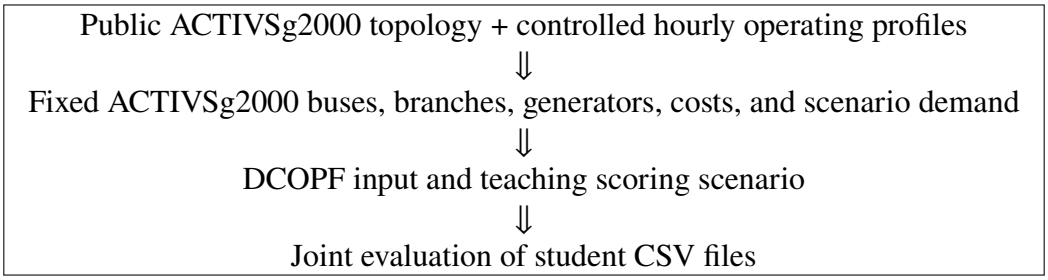
1.4 Learning objectives

- translate a planning idea into a machine-readable decision file;
- reason about spatial congestion, generator dispatch, and storage;
- understand why individually feasible plans can be jointly infeasible;
- use reproducible computational experiments and inspect diagnostics;
- communicate trade-offs among IT value, energy cost, and investment.

2 Data and model: what is real and what is instructional

The dispatch input is the public synthetic Texas ACTIVSg2000 model [2], which contains exactly **2,000 buses**, 3,206 branches, and 544 generator records. The fixed topology is combined with a controlled three-year hourly scenario package. The scenario provides deterministic bus-level load; active generator status, PMIN, and PMAX values come directly from the dataset. Because ACTIVSg2000 does not provide technology labels or hourly availability data, the active release does not apply synthetic wind/solar deratings. Each block is intended to be validated hour by hour with DCOPF. In the current repository environment, full validation is pending and unverified because the optional `highspy` dependency is unavailable; the checked-in manifests do not claim a passed release. It is an instructional scenario, not historical ERCOT telemetry or an official ERCOT operating model.

The conceptual pipeline is:



The following distinction is important:

- **Public data:** `bus`, `branch`, `generator`, and marginal-cost records from the ACTIVSg2000 source model.
- **Instructor preparation:** `formatting`, `validation`, and the full-horizon economic profiles supplied with the round.

- **Instructional assumptions:** the student-owned investments, connection requests, `DCOPF` treatment, `IT value`, capital-cost coefficients, storage-cost coefficients, and the `fair continuous projection` policy.

The exercise should not be interpreted as an ERCOT planning recommendation or as a forecast of actual interconnection feasibility.

3 Operating horizon and decisions

Each submission governs **one half-year with a one-hour time step:**

Year	Round	Hours	Zero-based period indices
2028	January–June	4,368	$0, \ldots, 4367$
2028	July–December	4,416	$0, \ldots, 4415$
2027 or 2029	January–June	4,344	$0, \ldots, 4343$
2027 or 2029	July–December	4,416	$0, \ldots, 4415$

The source year contains one full calendar year's hourly records. A submission therefore has access to roughly 2,000 possible buses and roughly 4,400 hourly evaluation states, but it does *not* encode those states as CSV rows. The canonical CSV lists only the sites used by your team; the evaluator combines those site decisions with the selected hourly scenario and economic artifacts.

3.1 Three-week, six-submission schedule

Week	Submission 1	Submission 2	Treatment
1	`2027_jan_jun`	`2027_jul_dec`	Feedback-only practice week; excluded from final ranking.
2	`2028_jan_jun`	`2028_jul_dec`	Scored weekly match; Submission 2 is the scored result.
3	`2029_jan_jun`	`2029_jul_dec`	Scored weekly match; Submission 2 is the scored result.

Within a week, the evaluator carries accepted sites, evaluator-assigned generator and storage assets, and the final storage state of charge from Submission 1 into Submission 2. Capital cost for an asset is charged once when it is first built in that weekly match. At the beginning of the next week, the asset ledger and SOC reset; no asset or SOC carries between weeks.

For each selected site s, submit a nonnegative connection capacity and optional private generation and storage capacities. A team's resources serve only that team. They cannot export energy to another team or become a shared pool. The `bus_id` must come from the supplied bus catalog.

4 Mathematical model

4.1 Connection load

Let $C_{i,s} \geq 0$ be the connection capacity submitted by team i for site s. The canonical CSV stores $C_{i,s}$ directly, together with optional generator and storage capacities; it does not store an hourly load profile. The evaluator uses the site-level plan when constructing each hourly dispatch.

4.2 Private generation

Each positive generator capacity in the canonical CSV represents a team-built generator. The seven-column CSV has no generator- or storage-ID columns; the evaluator assigns deterministic private asset IDs internally from the team and site IDs. These IDs are not lookup keys into the public generator catalog. The evaluator applies the same convex piecewise-linear operating-cost curve to every team-built generator at every site: the first 25% of capacity costs \$60/MWh, the next 50% costs \$100/MWh, and the final 25% costs \$160/MWh. You submit the capacity and the evaluator chooses hourly dispatch. For a generator with capacity $G_{i,g}$, dispatch satisfies

$$0 \le g_{i,g,t} \le G_{i,g}.$$

The generator is physically attached to the submitted bus and is private to your team. It is not an existing public generator and does not inherit a public generator ID, PMIN, PMAX, or marginal cost.

4.3 Private storage

For storage device b, submit power capacity $P_{i,b}$ in MW and energy capacity $B_{i,b}$ in MWh. The evaluator assigns the device's deterministic private asset ID internally; the student CSV contains no `storage_id` field. Charging $c_{i,b,t}$ and discharging $q_{i,b,t}$ are continuous. We assume ideal charging/discharging efficiency:

$$e_{i,b,t} = e_{i,b,t-1} + c_{i,b,t} - q_{i,b,t},$$

with

$$0 \le e_{i,b,t} \le B_{i,b}, \qquad 0 \le c_{i,b,t}, q_{i,b,t} \le P_{i,b},$$

$$q_{i,b,t} \le e_{i,b,t-1}, \qquad e_{i,b,0} = 0.5B_{i,b}, \qquad e_{i,b,H} \ge 0.5B_{i,b}.$$

Students submit storage capacities only. The grid operator chooses charging and discharging in the centralized dispatch. Storage is private to the submitting team: charging appears as additional team demand and discharging reduces that team's grid purchase; it cannot serve any other team's load. The evaluator assigns the storage asset ID from the team/site identity.

Storage cost calibration. The active half-year scoring files use the following instructional coefficients:

$$C_{i,b}^{\text{storage,cap}} = (\$350{,}000/\,\text{MW})P_{i,b} + (\$200{,}000/\,\text{MWh})B_{i,b}.$$

For a four-hour device, the two capital terms correspond to approximately \$1.15m/MW of installed power capacity. Storage operating cost is not charged. These are rounded teaching coefficients informed by U.S. battery-storage construction and O&M benchmarks, not a project quotation or an ERCOT tariff [12, 10, 9]. The four-hour relation gives a total capital equivalent of \$1.15m/MW. Storage operating cost is zero in the active evaluator.

4.4 DCOPF

For each hour, the instructor solves a linear DCOPF on the prepared Texas network. Network balance, generator limits, branch thermal limits, fixed base demand, and the centralized team-dedicated storage model are enforced. Load is inelastic in the ordinary feasible case: there is no integer unit commitment and no ordinary load-shedding variable. The solver, not the student, selects the storage schedule.

5 CSV submission format

Submit one UTF-8 CSV file per team. The canonical header is exactly:

```
team_id,site_id,bus_id,connection_capacity_mw,generator_capacity_mw,
    storage_power_mw,storage_energy_mwh
```

Each data row is one site. The file has no `period`, hourly `load_mw`, `submission_id`, `generator_id`, or `storage_id` columns and contains no hourly horizon rows. The `site_id` is a team-defined site label; `bus_id` must come from the supplied scenario block.

Example:

```
team_id,site_id,bus_id,connection_capacity_mw,generator_capacity_mw,
    storage_power_mw,storage_energy_mwh
team-blue,site-north,1001,40,20,10,40
team-blue,site-south,1002,15,0,0,0
```

Values use MW, MWh, and decimal notation. The `validator` checks the exact seven-column header, site identifiers, scenario bus identifiers, finite nonnegative capacities, and the storage ratio.

Before submitting, run the provided CSV `validator`. From the project root, run with the released ACTIVSg2000 scenario-block path:

```
PYTHONPATH=src python scripts/check_submission.py \\
  --file team-blue.csv
```

The `checker` is also available after installation as `ercot-game-check-csv`. Add `--artifact-dir` when you want the checker to verify scenario bus IDs; `--year`, `--round`, and `--horizon` are optional historical/report context. The validator is a convenience check; the instructor evaluator remains authoritative.

Historical compatibility note. The retired ten-column hourly format with `period` and `load_mw` is retained only for explicitly marked legacy fixtures and compatibility tests. Do not use it for a current submission. The legacy format's asset-ID fields are historical compatibility fields, not part of the active student contract.

5.1 Complete small example

The following is a complete canonical two-site example. It is intentionally small for format testing; a course submission may contain more site rows but still never contains hourly horizon rows:

```
team_id,site_id,bus_id,connection_capacity_mw,generator_capacity_mw,
   storage_power_mw,storage_energy_mwh
team-demo,site-north,1001,4,2,1,4
team-demo,site-south,1002,2,0,0,0
```

The second site has no generation or storage. The half-year hourly scenario is selected by the instructor and is not expanded into repeated rows in this file.

5.2 Checker troubleshooting

- **Unknown bus:** copy the `bus_id` from the supplied ACTIVSg2000 bus catalog.
- **Duplicate site:** give every site row a distinct `site_id`.
- **Storage ratio:** use storage energy equal to four times storage power, or set both storage fields to zero.
- **Malformed CSV:** save as UTF-8 CSV and inspect commas, quotation marks, and the exact header.

6 How joint evaluation works

The instructor first rejects `malformed CSV` files in isolation; a malformed file does not prevent other teams from being evaluated. Review the exact file contract in Section 5 before diagnosing a rejection.

Valid plans are then evaluated jointly. If the joint plan is `feasible`, each team is scored using its cleared hourly load, private resource dispatch, and the announced scoring vectors.

If the joint plan is `infeasible`, the evaluator applies the `fair continuous projection`, discussed in Section 7. No first-come-first-serve ordering is used. The projected load is then used for the score. The capital costs associated with submitted sites, generators, and storage remain part of the score. See Section 8 for the score calculation.

7 Fair continuous projection

If the valid teams' full request is jointly infeasible, the evaluator applies fixed anti-manipulation caps before searching for a projected dispatch. Each site is capped at 100 MW, each team at 500 MW, and in a multi-team case the largest team request is capped at the smallest positive team request. These caps are instructor-side policy; students cannot tune them.

Let $d_{i,s,t}$ be the capped requested load of team i. The evaluator chooses a separate service fraction $\alpha_i \in [0, 1]$ for each team and evaluates $\widehat{d}_{i,s,t} = \alpha_i d_{i,s,t}$. It first maximizes total served load. It then restricts the search to solutions within $\max(10^{-6}, 10^{-6}L^*)$ MWh of that optimum L^* and maximizes

$$\sum_{i:d_i>0} \log(\alpha_i d_i),$$

which is proportional-fair tie-breaking. Positive valid requests have a minimum service fraction of 10^{-6}. The result records each team's fraction, requested load, projected load, and the cap policy.

This policy keeps valid teams eligible, prevents an oversized request from dominating the projection, and produces a physically feasible joint dispatch. It is a teaching rule, not an ERCOT operating procedure [7, 5, 3].

8 Scoring

For team i, the `score` is

$$S_i = V_i^{\mathrm{IT}} - C_i^{\mathrm{energy}} - C_i^{\mathrm{gen,op}} - C_i^{\mathrm{gen,cap}} - C_i^{\mathrm{storage,cap}} - C_i^{\mathrm{site}}.$$

The components are:

- V_i^{IT}: cleared load multiplied by the explicit full-horizon `IT value` vector. The active AI-intensive profile uses \$1500/MWh at every hour;
- C_i^{energy}: for Submission 2, grid purchase multiplied by the nonnegative LMP from the selected joint dispatch at the selected bus and hour. Submission 1 has no energy-purchase cost;
- $C_i^{\mathrm{gen,op}}$: team-generator dispatch charged at the common 25%/50%/25% curve of \$60/100/160 per MWh;
- $C_i^{\mathrm{gen,cap}}$: newly built generator MW multiplied by \$1.3m/MW, charged once per weekly match;
- $C_i^{\mathrm{storage,cap}}$: newly built storage charged at \$350,000/MW plus \$200,000/MWh, with no storage operating cost;
- C_i^{site}: newly built connection capacity multiplied by \$20m/MW, charged once per weekly match;

The active instructional coefficients are listed below. The selected rule file and economic-profile artifacts remain authoritative for a released round.

Parameter	Active value	Basis
Generator capital	$1.3m/MW	One-time weekly-match capital coefficient; informed by EIA/NREL technology-cost methods.
Site cost	$20m/MW	Capacity-dependent instructional connection coefficient, informed by JLL shell/core and AI fit-out costs [6].
Generator operating cost	Piecewise $60/100/160 per MWh at 25/50/25%	Common team-built curve; EIA heat-rate and generator-cost references inform the mapping [11, 13].
Energy price	Submission 2 joint-dispatch LMP	Locational price at the selected bus and hour; Submission 1 has no energy-purchase cost.
IT value	$1500/MWh hourly	Full-horizon AI-intensive opportunity value; cloud pricing pages are contextual references, not a direct tariff conversion [1, 8, 4].
Storage power capital	$350,000/MW	Current power-capacity coefficient.
Storage energy capital	$200,000/MWh	Current energy-capacity coefficient; together with power capital gives $1.15m/MW for a four-hour system.
Storage operation	Not charged	The active evaluator has no storage operating-cost term.
Invalid file score	0	Deterministic grading policy, not a market quantity.

8.1 How to reason about your score

The evaluator first determines how much of your requested load is served. It then adds IT value for served load and subtracts energy, site, generator, and storage costs. A larger plan is therefore not automatically better: an additional site may increase IT value while also increasing energy, congestion, and capital costs. A generator or storage device can reduce grid purchases but still lower the final score if its costs exceed the avoided energy cost.

Use the full-horizon `IT-value` vector and the locational `price` artifact supplied with the round when estimating a score. The current IT-value vector is constant by policy; the energy price is not a student-chosen value.

9 Practical workflow

1. Read the weekly schedule and identify the evaluator's scenario block.
2. Inspect the ACTIVSg2000 bus and generator tables supplied by the instructor.
3. Choose one or more buses and specify site capacities in the canonical CSV.

4. Decide whether private generation or storage improves your score.
5. Generate the `CSV` and run the `validator`.
6. Keep the exact submitted file and validator output for reproducibility.
7. Submit both weekly files before the announced deadlines; revise the first-stage plan for Submission 2 if the instructor permits revisions.

9.1 What to inspect after evaluation

Read the result in this order:

1. feasibility or projected service status and your team-specific service fraction;
2. total score and rank;
3. cleared load, grid purchases, and resource costs;
4. binding or limiting network diagnostics;
5. storage state and generator dispatch;
6. raw audit output only when debugging.

9.2 Final submission checklist

Before uploading, confirm:

- the file is `UTF-8 CSV` with the exact header;
- each data row has a distinct site ID and the exact seven-column header;
- all bus IDs come from the supplied catalog;
- all capacities are finite and nonnegative, with storage energy equal to four times storage power when storage is present;
- the `checker` passes without requiring hourly context arguments;
- you retained the exact submitted file and checker output;
- your strategy note explains the main economic and physical trade-offs.

10 Troubleshooting and frequently asked questions

How are generator and storage assets identified?
The active seven-column CSV has no asset-ID columns, including no generator-ID or storage-ID columns. Submit capacities only; the evaluator assigns deterministic private asset IDs from the team and site IDs. You cannot choose the operating-cost curve: every team-built generator uses the common piecewise curve specified in Section 8, and the evaluator dispatches it.

Can my generator or storage serve another team?
No. The grid operator may schedule your storage, but storage discharge and team-generator output can reduce only your team's load. They cannot be shared with another team.

What happens if the joint plan is infeasible?
The evaluator applies the capped fair projection in Section 7, using a separate service fraction for each valid team. The projection first maximizes total service and then uses proportional-fair tie-breaking.

What carries from Submission 1 to Submission 2?
Accepted sites, evaluator-assigned generators and storage assets, and final storage SOC persist within the week. Capital is charged once per weekly match. The ledger and SOC reset at the start of the next week.

Can I schedule storage myself?
No. Submit only storage power and energy capacities. The centralized grid optimizer schedules charge, discharge, and SOC subject to the four-hour, team-dedicated constraints.

Why is my score negative?
Served-load value may be smaller than energy, site, generator, or storage costs. Inspect the detailed score components before changing the plan.

Which result is official?
The instructor-generated result and audit files determine the official score. Week 1 is feedback-only; final ranking uses scored Submission 2 results from Weeks 2 and 3. Local calculations are for debugging and strategy development.

Why can the instructor not release a score yet?
The checked-in scenario manifests are currently unverified because the optional `highspy` dependency is unavailable. The instructor must run complete HiGHS-backed validation before classroom use. The Colab benchmark also requires `highspy`, `psutil`, prepared artifacts, and a compatible runtime; an unrun benchmark cannot justify shortening the official half-year horizon.

11 References

References

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